Gunnar König
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Personal Data

Work	Maria-von-Linden-Straße 6, 72076 Tübingen, Germany
Details	Born 1994 in Reutlingen, Germany. Citizenship: German

Academic Positions

2023–present	**Postdoctoral Researcher**, Tübingen AI Center, University of Tübingen Supervisor: Prof. Ulrike von Luxburg, Theory of Machine Learning Group. Topic: Explainable AI.

Education

2019–23	**PhD in Machine Learning**, MCML, LMU Munich Supervisors: Prof. Bischl (LMU), Prof. Grosse-Wentrup (University of Vienna) Thesis: "If interpretability is the answer, what is the question?" Committee: Prof. Bischl (LMU), Prof. Augustin (LMU), Prof. Lipton (CMU). Topics: Explainable AI, Causal Inference.
2016–18	**M.Sc. in Data Science**, Elitenetzwerk Bayern, LMU Munich Thesis: "Causal Relationships in Calcium Imaging" Topics: Causal Inference, Deep Learning, Medical Data.
2015	**Exchange semester**, École Polytechnique de Montréal Topics: Deep Learning, Biomedical Data Analysis.
2012–16	**B.Sc. in Information Systems**, TU Munich Thesis: "Online Classification Using Adaptive Sparse Grid Kernels" Topics: Machine Learning, Algorithms, Entrepreneurship. Additional coursework from **B.Sc. Maths** (2015–16).
2012	**Abitur**, Friedrich-List-Gymnasium Reutlingen Awards for outstanding grades in maths, physics, chemistry, and for the overall grade (1.0). Special award for social outreach against discrimination.

Publications

760+ citations (Google Scholar), h-index: 9. The 5 most important papers are highlighted **bold**.

G. König*, E. Günther*, and U. von Luxburg. **Disentangling Interactions and Dependencies in Feature Attribution**. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024a.

F. K. Ewald, L. Bothmann, M. N. Wright, B. Bischl, G. Casalicchio, and G. König. **A Guide to Feature Importance Methods for Scientific Inference**. In *World Conference on Explainable Artificial Intelligence (XAI)*, 2024.

G. König*, S. Dandl*, K. Blesch*, T. Freiesleben*, J. Kapar, B. Bischl, and M. N. Wright. Counterfactuals – generating plausible model-agnostic counterfactual explanations with adversarial random forests. In *World Conference on Explainable Artificial Intelligence (XAI)*, 2024b.

T. Freiesleben, G. König, C. Molnar, and A. Tejero-Cantero. Scientific inference with interpretable machine learning: Analyzing models to learn about real-world phenomena, journal = Minds and Machines, volume = 34, number = 3, pages = 32, year = 2024.

C. Molnar, G. König, B. Bischl, and G. Casalicchio. Model-agnostic feature importance and effects with dependent features: A conditional subgroup approach. *Data Mining and Knowledge Discovery*, 38(5):2903–2941, 2024.

G. König, T. Freiesleben, and M. Grosse-Wentrup. **Improvement-Focused Causal Recourse (ICR)**. In *AAAI Conference on Artificial Intelligence (AAAI), oral presentation*, 2023.

G. König*, C. Luther*, and M. Grosse-Wentrup. **Efficient SAGE Estimation via Causal Structure Learning**. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2023a.

T. Freiesleben and G. König. Dear XAI community, we need to talk: Fundamental misconceptions in current XAI research. In *World Conference on Explainable Artificial Intelligence (XAI)*, 2023.

G. König. *If Interpretability Is the Answer, What Is the Question?* PhD thesis, Ludwig-Maximilians-Universität München, 2023.

G. König*, C. Molnar*, T. Freiesleben*, J. Herbinger, T. Reisinger, G. Casalicchio, M. N. Wright, and B. Bischl. Relating the partial dependence plot and permutation feature importance to the data-generating process. In *World Conference on Explainable Artificial Intelligence (XAI)*, 2023b.

G. König, C. Molnar, B. Bischl, and M. Grosse-Wentrup. **Relative Feature Importance**. In *International Conference on Pattern Recognition (ICPR)*, 2021a.

G. König, T. Freiesleben, and M. Grosse-Wentrup. A causal perspective on meaningful and robust algorithmic recourse. In *ICML Workshop on Algorithmic Recourse*, 2021b.

C. Molnar, G. König, J. Herbinger, T. Freiesleben, S. Dandl, C. A. Scholbeck, G. Casalicchio, M. Grosse-Wentrup, and B. Bischl. General pitfalls of model-agnostic interpretation methods for machine learning models. In *International Workshop on Extending Explainable AI Beyond Deep Models and Classifiers*, 2020.

Selected Talks

2025 **Disentangling Interactions and Dependencies in Feature Attribution**
Theory of Interpretable AI Seminar, Host: Tim van Erven (University of Amsterdam)

2025 **Disentangling Interactions and Dependencies in Feature Attribution**
Interpretable Machine Learning Seminar, Online, Hosts: Giuseppe Casallicchio (LMU Munich) and Marvin Wright (BIPS Bremen)

2025 **Disentangling Interactions and Dependencies in Feature Attribution**
Stochastics Seminar, University of Amsterdam, Host: Tim van Erven (University of Amsterdam)

2023 **Improvement-focused Causal Recourse**
Research Seminar Machine Learning Theory, Host: Ulrike von Luxburg (University of Tübingen)

2023 **Improvement-focused Causal Recourse**
AAAI Conference, Oral Presentation

2022 **Relative Feature Importance**
Mind Team Meeting, Host: Bertrand Thirion (Inria Saclay Paris)

2021 **Interpretable Machine Learning – Jack of All Trades, Master of None?**
Convey Seminar, Host: Debarghya Ghoshdastidar (TU Munich)

Teaching

2024 **Explainable Machine Learning**, Universtiy of Tübingen
Seminar, 3 ECTS, 20 students, Master level; main instructor.
Together with Ulrike von Luxburg, Sebastian Bordt, Robi Bhattacharjee.

2022 **Multivariate Statistics**, LMU Munich
Exercise, 2 ECTS, 20 students, Master; tutor and exam.
Together with Giuseppe Casalicchio

2022 **Advanced Machine Learning**, LMU Munich
Lecture and Exericse, 6+3 ECTS, Master; lecture material and exercises.
Together with Giuseppe Casalicchio, Viktor Bengs, Susanne Dandl

2022 **Introduction to Causal Inference**, LMU Munich
Seminar, 6 ECTS, Bachelor; main instructor.
Together with Fabian Scheipl, Susanne Dandl, Moritz Herrmann

2021 **Explainability and Causality**, Austrian Academy of Science, Vienna
2 day tutorial, PhD students; main instructor
Together with Moritz Grosse-Wentrup

2021 **Tutorial Explainability and Causality**, ZENITH doctoral programme, Online
2 day tutorial, PhD students; exercise
Together with Moritz Grosse-Wentrup

2021 **Philosophy and Explainable AI**, LMU Munich
Seminar, 6-9 ECTS, 30 students, Master; main instructor
Together with Timo Freiesleben

2021 **Seminar Causality and Graphical Models**, LMU Munich
Seminar, 6-9 ECTS, 15 students, Master; main instructor
Together with Susanne Dandl, Timo Freiesleben

2021 **Current Reseach in Data Science**,LMU Munich
Seminar, 9 ECTS, 20 students, Master
Together with Giuseppe Casalicchio

2020 **Seminar Ethics in AI**, LMU Munich
Seminar, 6-9 ECTS, 15 students, Bachelor and Master; main instructor
Together with Susanne Dandl, Christoph Molnar, Florian Pfisterer

2020 **Seminar Current Research and Applications in Interpretable Machine Learning**, LMU Munich
Seminar, 9 ECTS, 20 students, Master
Together with Giuseppe Casalicchio, Christoph Molnar, Julia Herbinger

2020 **Seminar Causality and Graphical Models**, LMU Munich
Seminar, 6-9 ECTS, 10 students, Master; main instructor
Together with Christoph Molnar, Susanne Dandl

2019 **Seminar Limitations of Interpretable Machine Learning**, LMU Munich
Seminar, 6-9 ECTS, 10 students, Master
Together with Christoph Molnar, Giuseppe Casalicchio, Christian Scholbeck, Florian Pfisterer

Supervision

2024-25 **Eric Günther**, PhD Student
Topic: Disentangling Interactions and Dependencies in Feature Attribution, published at AISTATS 2025

2024 **Anna Vollweiter**, Master Thesis
Topic: Long-term Robustness of Recourse

2024 **Selina Mail**, Bachelor Thesis
Topic: Feature Importance for Conditional Independence Testing

2023-24 **Fiona Ewald**, PhD Student
Topic: A Guide To Feature Importance Methods for Scientific Inference, published at XAI 2024

2023 **Rebekka Schade**, Bachelor Thesis
Topic: Algorithmic Recourse in Practice

2021-23 **Christoph Luther**, Master Thesis, PhD Student
Topic: Efficient SAGE Estimation via Causal Structure Learning, published at AISTATS 2023

2020 **Valentyn Melnychuck**, Master Thesis
Topic: Conditional Normalizing Flows for Feature Importance

Non-academic Positions

2017–23 **Freelance Data Scientist**
Led and grew an international team of data scientists to build a deep learning based disease diagnosis system for a Singapore-based startup.
Gave lectures introducing ML fundamentals to industry practioners.

2016 **Data Scientist**, Siemens, Munich
Using Machine Learning to predict financial indexes for in-house customers. The systems are now used in production.

Community Service And Social Outreach

2019–now **Reviewing** for AISTATS, FAccT, NeurIPS, TMLR, Nature.

2018–22 **Co-Founder and Vice President**, ThinkTech (prev. ConsciousCoders)
I helped start and grow the nonprofit to more than 30 members with computer science, law, politics, design, and philosophy backgrounds. Among others, we taught university lectures and created workshops for schools, industry, and policymakers. My highlights were the side events we hosted at the United Nations in Geneva to inform about the dangers of autonomous weapon systems.

2018–22 **Co-Founder**, WeRePack.org
Plugin to help online shops to reuse packaging and reduce waste. The initiative was founded as part of the SEA:certificate by the Social Entrepreneurship Academy and won the Strascheg Award.

Last updated: July 8, 2025